

Question 16**1 mark**

Identify 10° in radians.

a) $\frac{1800}{\pi}$

b) $\frac{\pi}{1800}$

c) $\frac{18}{\pi}$

d) $\frac{\pi}{18}$

Question 17**1 mark**

The polynomial function, $P(x) = a(x-1)^2(x+4)^2$, has a y -intercept of -8 .

Identify the value of a .

a) -2

b) $-\frac{1}{2}$

c) $\frac{1}{2}$

d) 2

Question 18**1 mark**

Identify the value of $\log_4\left(\frac{1}{16}\right)$.

- a) -2
- b) $-\frac{1}{2}$
- c) $\frac{1}{2}$
- d) 2

Question 19**1 mark**

Given the angle $\frac{25\pi}{7}$, identify the coterminal angle on the interval $[-2\pi, 0]$.

- a) $\frac{18\pi}{7}$
- b) $\frac{11\pi}{7}$
- c) $-\frac{3\pi}{7}$
- d) $-\frac{10\pi}{7}$

Question 20**1 mark**

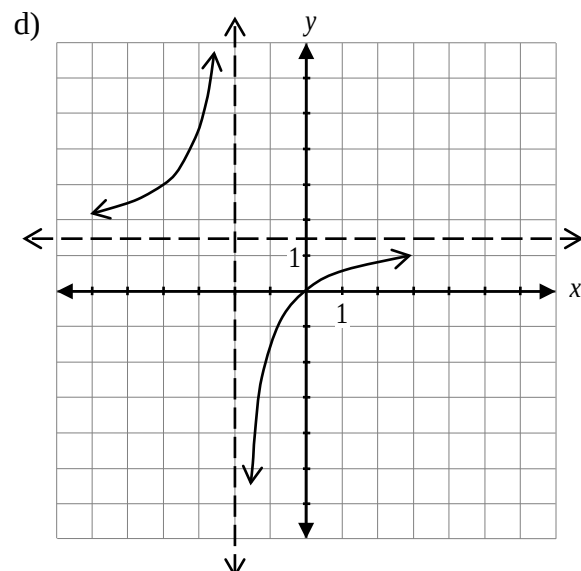
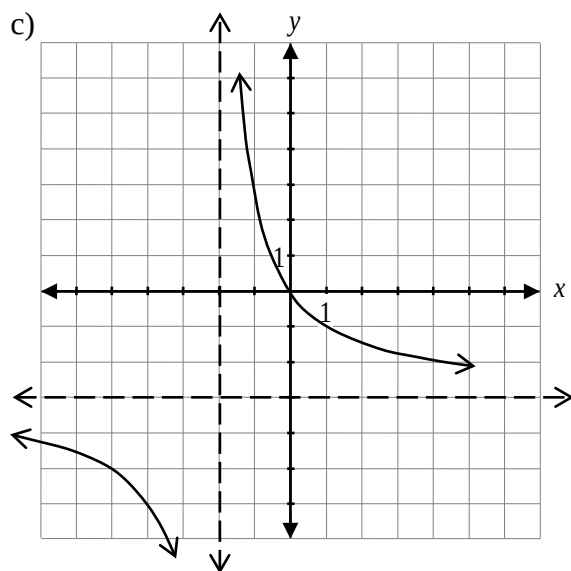
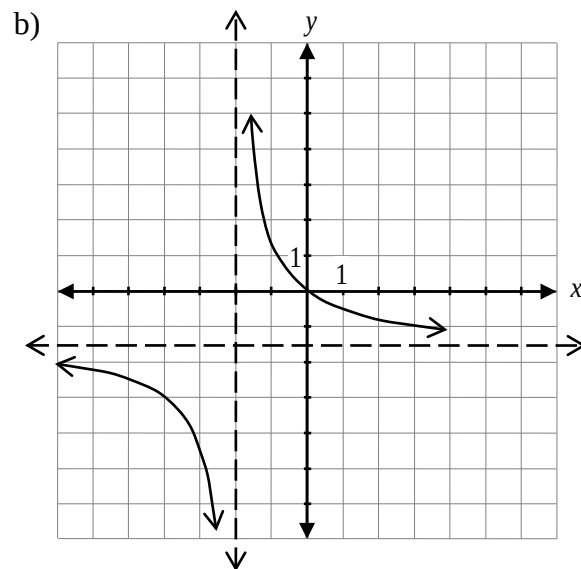
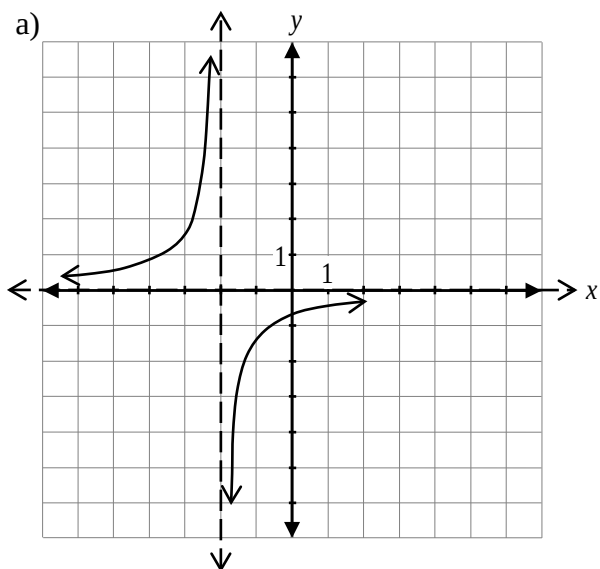
Identify which expression cannot be evaluated.

- a) ${}_7P_0$
- b) ${}_7P_6$
- c) ${}_7P_7$
- d) ${}_7P_8$

Question 21

1 mark

Identify the graph of $f(x) = \frac{-3x}{2x+4}$.



Question 22**1 mark**

Given a point $(-2, 0)$ on the graph of $y = f(x)$, identify the coordinates of the corresponding point on the graph of $y = 4f\left(\frac{1}{2}x\right)$.

- a) $(-8, 0)$
- b) $(-4, 0)$
- c) $(-2, 0)$
- d) $(-1, 0)$

Question 23**1 mark**

Identify the non-permissible value of θ for the expression $\frac{\cos \theta}{1 + \sin \theta}$.

- a) $\frac{\pi}{2}$
- b) π
- c) $\frac{3\pi}{2}$
- d) 2π

Question 24**1 mark**

Identify the function with an asymptote at $x = -3$.

- a) $y = \log(x + 3)$
- b) $y = \log x + 3$
- c) $y = \log(x - 3)$
- d) $y = \log x - 3$

Question 25

3 marks 117

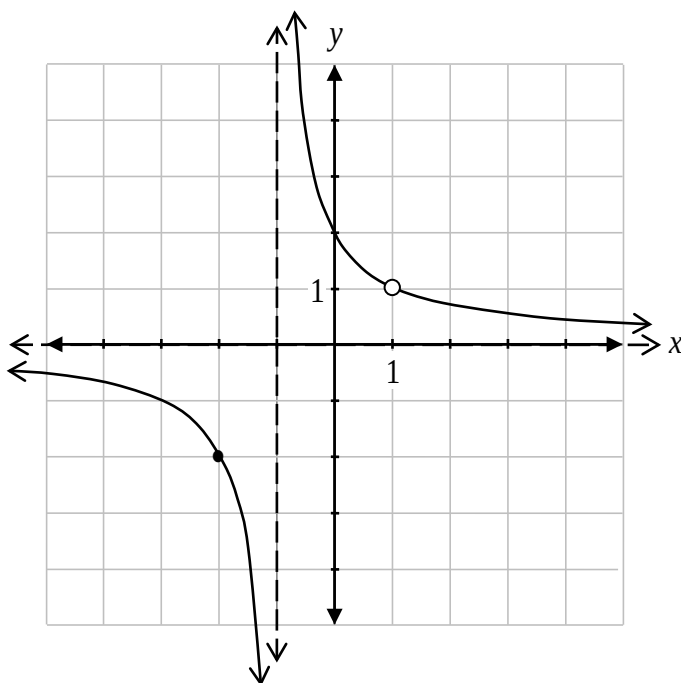
Evaluate the following expression.

$$\tan\left(\frac{2\pi}{3}\right)\csc\left(\frac{-2\pi}{3}\right) + \cos(3\pi)$$

Question 26

1 mark 118

State the range of the graph below.

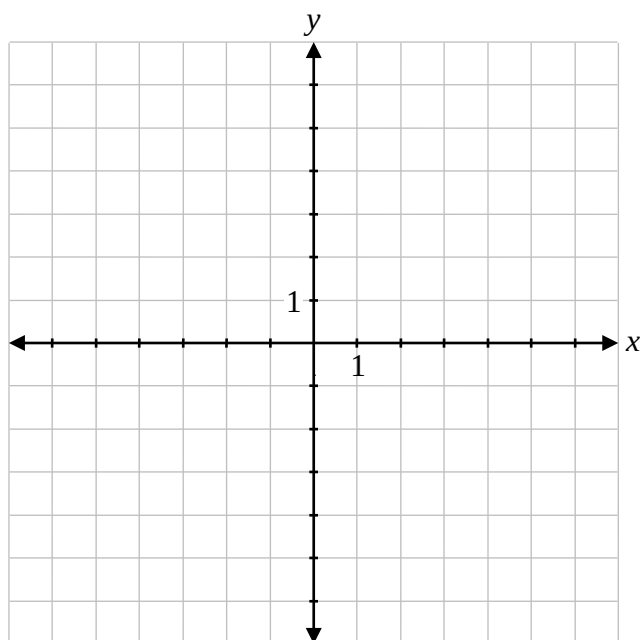


Range: _____

Question 27

2 marks 119

Sketch the graph of the function $f(x) = \frac{2x^2 - 5x}{x}$.



Question 28

1 mark 120

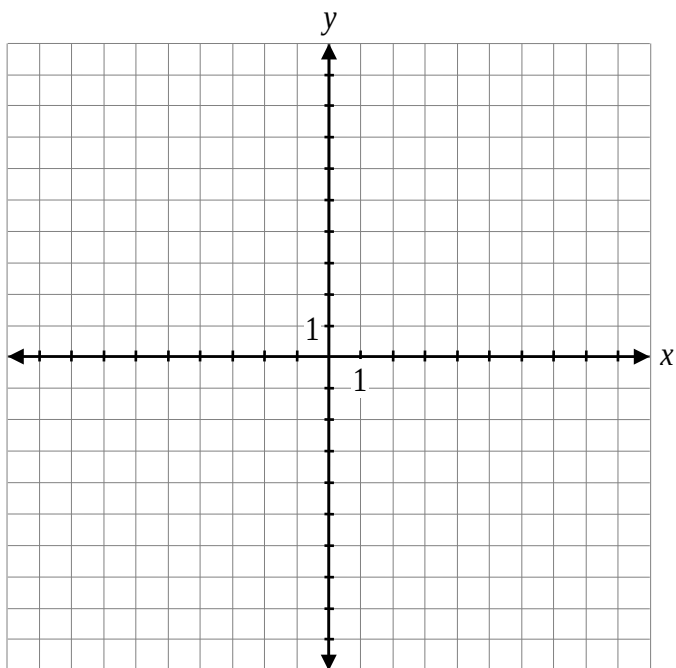
State a possible value of n if the polynomial function $P(x) = (x - 1)^2(x + 2)^n$ has a range of $[0, \infty)$.

Question 29

2 marks

121

Sketch the graph of $y = \left(\frac{1}{2}\right)^{x-1}$.



Question 30**1 mark**

122

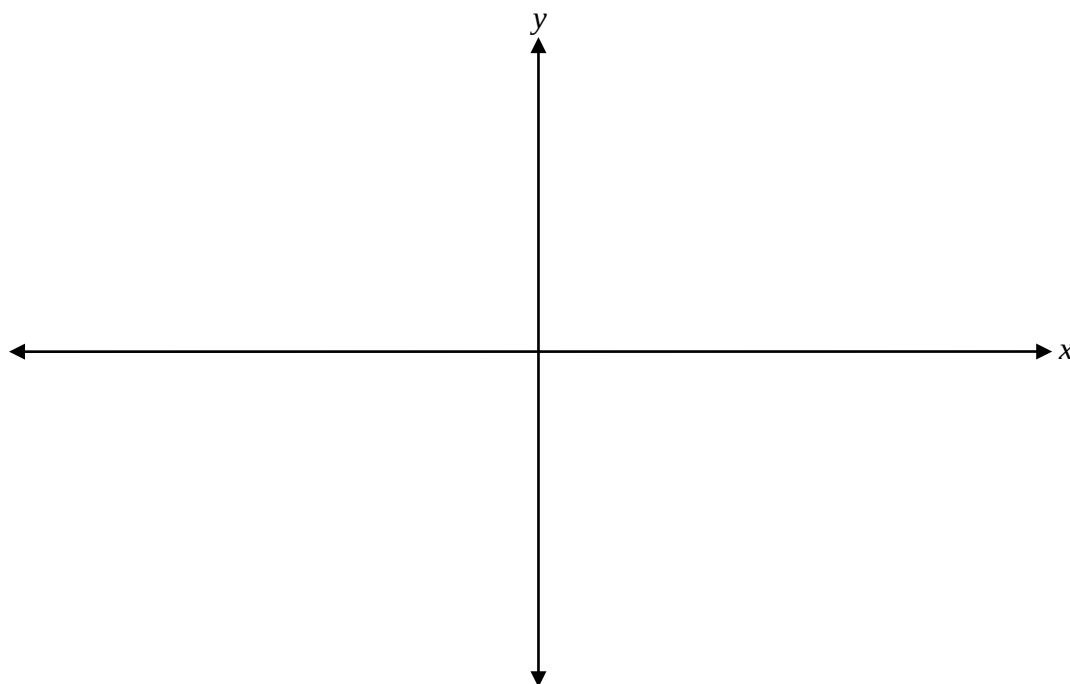
Solve.

$$\log_x 27 = 3$$

Question 31**2 marks**

123

Sketch at least two periods of the graph $y = \tan x$.

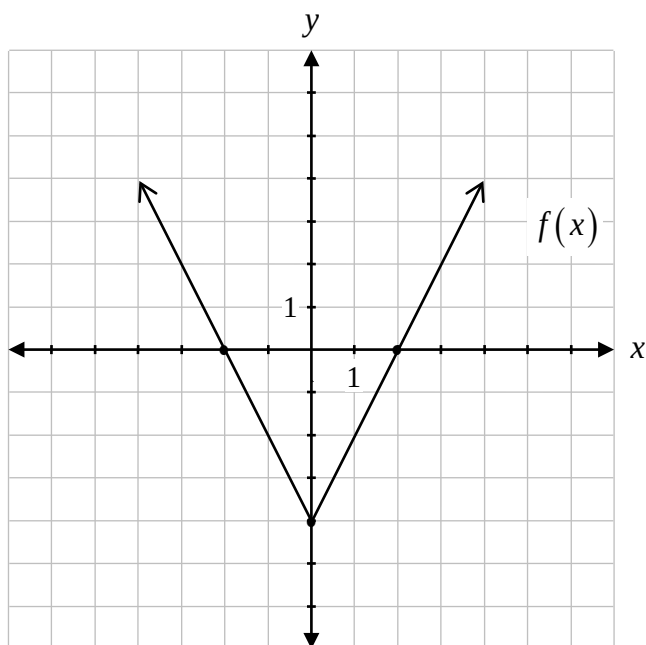


Question 32

1 mark

124

Given the graph of $f(x)$, determine the domain of $\frac{1}{f(x)}$.



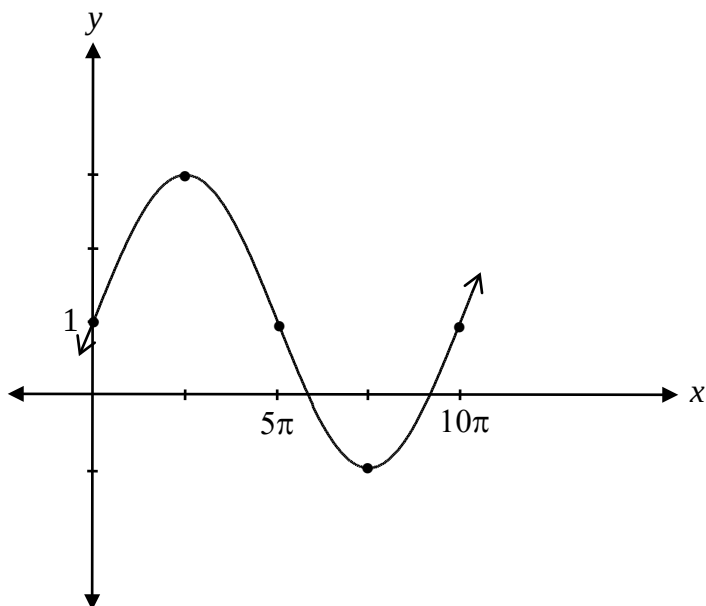
Domain: _____

Question 33

3 marks

125

Determine the values of A, B, and D of the sinusoidal function in the form $y = A \sin(Bx) + D$.



A = _____

B = _____

D = _____

Question 34

1 mark 126

Determine if the point $\left(-\frac{\sqrt{7}}{5}, \frac{2}{5}\right)$ is on the unit circle.

Justify your answer.

Question 35

3 marks 127

Solve, algebraically.

$$\frac{{}_nC_5}{{}_nC_4} = 6$$

Question 36

2 marks

128

Given $\sin \alpha = \frac{4}{5}$, where α is in quadrant II, determine the exact value of $\sin 2\alpha$.

Question 37

a) 1 mark b) 2 marks

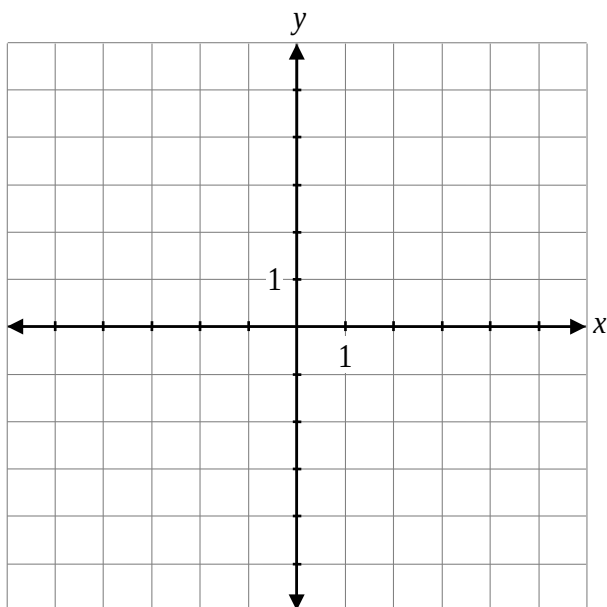
129
130

Given the functions $f(x) = x + 1$ and $g(x) = \sqrt{x}$,

a) determine the equation of $g(f(x))$.

$$g(f(x)) = \underline{\hspace{4cm}}$$

b) sketch the graph of $g(f(x))$.



Question 38

1 mark 131

Steve is asked to determine an equation with a larger period than the period of the graph of $y = \cos(2x)$.

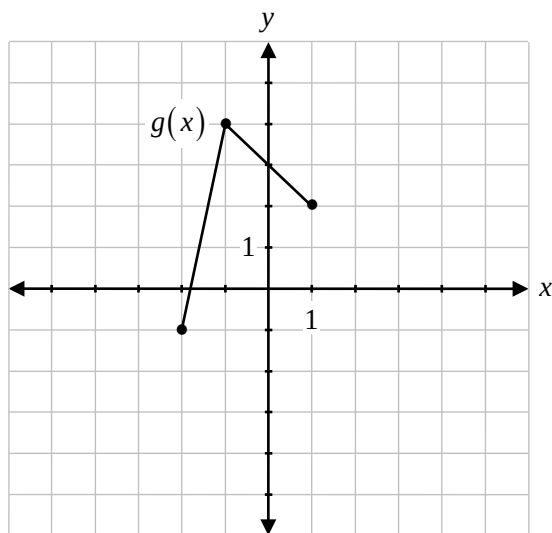
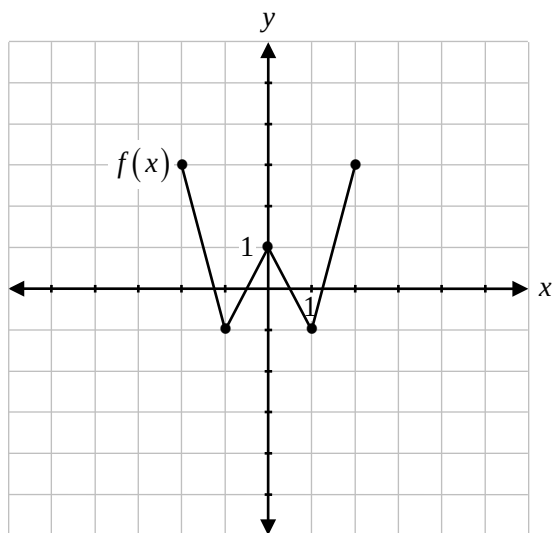
Justify why Steve's answer of $y = \cos(6x)$ is incorrect.

Question 39

a) 1 mark b) 1 mark

132
133

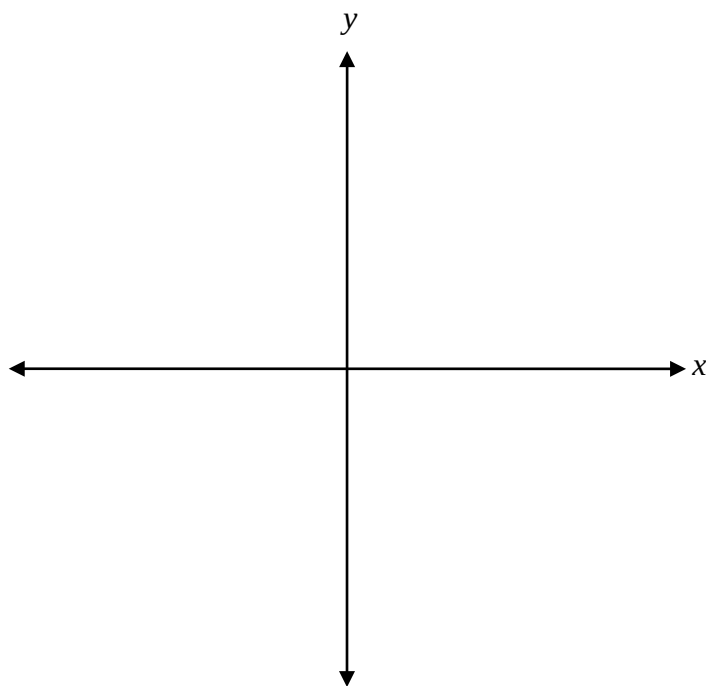
Given the graphs of $f(x)$ and $g(x)$,



a) determine the value of $(f \cdot g)(-1)$.

b) determine the value of $g(f(0))$.

Sketch the graph of $P(x) = -(x-1)^3(x-3)(x+1)$.



Question 41

a) 1 mark b) 1 mark

135
136

The point $(-\sqrt{3}, 1)$ is on the terminal arm of an angle θ , in standard position.

a) Determine $\tan \theta$.

b) Determine a possible value of θ , in radians.

Question 42**1 mark**

137

Describe the transformation used to obtain the graph of $y = \log_5 x$ given the graph of $y = 5^x$.

Question 43**2 marks**

138

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$, where $\theta \in \mathbb{R}$.

Question 44**1 mark** 139

Given that the point (a, b) is on the graph of $f(x)$, describe how you would determine the corresponding point on the graph of $y = \sqrt{f(x)}$.

Question 45**1 mark** 140

Evaluate.

$$\cos\left(\frac{\pi}{20}\right)\cos\left(\frac{\pi}{5}\right) - \sin\left(\frac{\pi}{20}\right)\sin\left(\frac{\pi}{5}\right)$$

Question 46**3 marks**

141

Describe the transformations used to obtain the graph of the function $y = f(-x + 6) - 8$ from the graph of $y = f(x)$.

Question 47**2 marks**

142

State the equations of all the asymptotes of the function, $y = \frac{1}{3x+1}$.

Question 48

3 marks

143

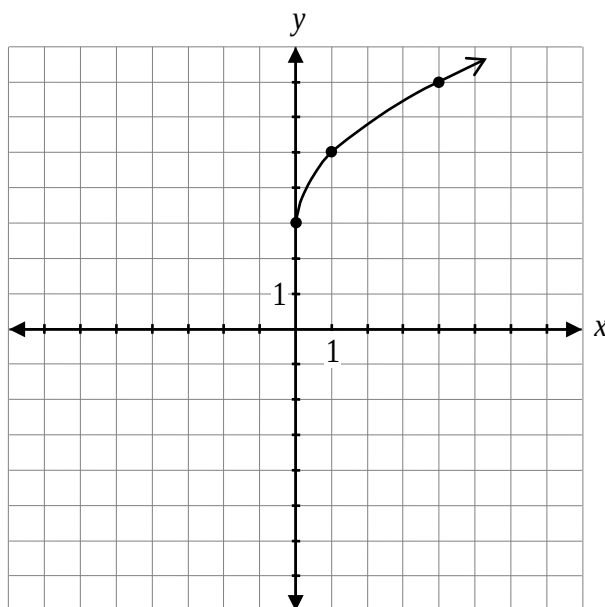
Determine the zeros of the polynomial function $P(x) = 2x^3 + 5x^2 - 4x - 3$.

Question 49

2 marks

144

Determine the equation of the radical function represented by the graph.



$y =$ _____

No marks will be awarded for work done on this page.

